HELLOLILLY · RESEARCH

What the evidence says

The research behind HelloLilly, verified, corrected, and mapped to every design decision.

RESEARCH COMPANION TO VERSION 4

DATE OCTOBER 2026 CONFIDENTIAL DRAFT

CONTENTS

What is inside

· Opening

01 Why People Get Stuck

02 What Actually Works

03 Digital Tools and AI

04 Re-skilling and Education

05 Who Is Hit, and How

06 How Big the Disruption Is

07 From Evidence to Design

08 What a Digital System with Coaches Can Carry

09 What We Should Measure

10 What Not to Cite

· Appendix: The Swedish Data Behind the System

A companion to the HelloLilly Jobseeker Product Vision, Version 4. Principles and boundaries are in Addendum A.

Opening

A system that promises to help people back to work owes them evidence, not enthusiasm. This document collects what the research actually says about why people get stuck, what helps, what does not, where digital tools and AI help and where they harm, and how re-skilling really pays off. It then shows how each finding shapes a specific part of the HelloLilly system.

Two things make it more than a literature list. First, every number has been reconciled against an adversarial verification pass: several widely repeated figures were found to be wrong, mis-attributed or retired, and they are listed at the end so nobody repeats them. Second, every finding is tied to a design consequence in the Version 4 system, so the research is not decoration. It is the reason the system looks the way it does.

How to read the confidence levels

LEVEL	MEANING
Strong	Meta-analysis, randomised trial, or large register study, confirmed across independent passes
Good	Peer-reviewed or authoritative source, verified, sometimes with a correction
Indicative	Single source, observational, institutional report or preprint: shapes design, never quoted as proof
Projection	Forecasts and automatability estimates: not observed outcomes

Most of the causal backbone is quasi-experimental and meta-analytic, reinforced by a few narrow randomised trials. That is a strong base for honest design, and a weak base for grand promises.

SECTION 01

Why People Get Stuck

01

Unemployment harms mental health, causally

Strong. Across hundreds of studies, unemployed people report substantially worse mental health than employed people (an average difference around d = 0.5), and about one in three unemployed people show psychological problems against one in six of the employed. Longitudinal studies and natural experiments indicate that unemployment causes the harm rather than merely accompanying it, and that re-employment improves it, though not always fully.

Part of the harm is the loss of what work gives besides money: time structure, social contact, purpose, status and activity.

For HelloLilly: the system must give structure and contact back, not only information. That is the job of Progress Support, the Community and the coach. And the tone must never add to the burden: argue for the person, never judge them.

Job search is a self-regulation task

Strong. Job search requires setting goals, planning, sustaining effort, reading feedback and coping with setbacks over months. Self-efficacy, clear goals, planning and search intensity predict offers and re-employment; negative affect and poor coping predict slower return.

For HelloLilly: one clear next step at a time, a living plan, visible progress, and feedback that explains rather than discourages.

The practical barriers

Good. Recurring obstacles are poor applications, weak tailoring, weak interview skills, not knowing which channels work, and too narrow or too unfocused a search. High volumes of untargeted applications do not work; they increase rejection and discouragement. Better targeting does.

For HelloLilly: fewer, better applications. Match Analysis, the Go or Cut gate, the tailored CV, Interview Preparation and the Contact Planner all exist to replace volume with quality.

Disadvantage accumulates with time out of work

Good. The longer someone is out of work, the more mental health, search effort and employer interest decline, and the more learned helplessness sets in. Speed matters.

For HelloLilly: restore momentum early; make the first week count.

The pains specific to changing career

Indicative. Career changers face a distinct layer: an occupational identity they are reluctant to give up, decision paralysis when there is no target role, the financial risk of starting over, loss of status, fatigue at starting credentials from scratch, and imposter feelings. The evidence here is mostly qualitative, so these pains are designed for, but no effect sizes are attached.

For HelloLilly: the Transition Compass, Direction Explorer, Prior Learning Check and Test Before You Invest exist for exactly these pains.

SECTION 02

What Actually Works

02

Job-search help works, when it builds both skill and motivation

Strong. A meta-analysis of 47 studies found that job-search interventions raise the odds of employment about 2.67 times, but **only** when they build both job-search skills and motivation. The effect runs through skills, self-efficacy and actual search behaviour. Skills-only or motivation-only programmes underperform.

For HelloLilly: this is the case for the division of labour. The system carries skill and execution; the coach carries motivation. Neither alone is enough.

Job clubs work

Strong. Structured job clubs, with daily tasks, peer support, practice and accountability, placed more people faster than unstructured search. Effects are larger where structure and active practice are kept high.

For HelloLilly: the Community, peer support in the Expertise Exchange, Progress Support and Interview Training emulate the job-club ingredients.

Building mastery works, most for those starting lowest

Good. The JOBS II randomised trial found significantly higher re-employment, higher income and better mental health two years on, with the largest benefits for people who started with low motivation and low sense of mastery. (The often-quoted "55 per cent higher odds" could not be verified and must not be cited.)

For HelloLilly: confidence is not soft; it is causal. Interview Training, rehearsal and being accompanied through the work are part of the outcome, not decoration.

Place first, then support

Strong, with a population caveat. Individual placement and support, which places people quickly into real work and supports them afterwards, raises competitive employment considerably for people with serious mental illness (in a Swedish trial, 46 per cent against 11 per cent at eighteen months). The numbers do not transfer to all jobseekers; the mechanism does: rapid placement with in-work support beats train-then-place.

For HelloLilly: the Placement Finder, the Employer Bridge and the fork that always shows "apply now".

Programme type predicts effect

Good. Job-search assistance gives earlier gains; human-capital training pays off later; subsidised public-sector jobs are ineffective; information-only measures such as one-off seminars or leaflets have limited impact. Small if-then planning prompts help follow-through, but never replace substantive support.

For HelloLilly: information alone is not help. Every explanation in the Situation Guide ends in a next step, and Lilly hands over to action, not to reading.

Coaching works when it is supportive, not when it is surveillance

Good. More frequent meetings with clear expectations and supportive counselling improve job-finding; purely administrative monitoring can feel punitive and works less well.

For HelloLilly: support, not surveillance. Activity is logged to help the person, never to monitor them.

SECTION 03

Digital Tools and AI

03

Better recommendations change behaviour, not necessarily outcomes

Strong. Large field experiments with algorithmic job recommendations, including one on Sweden's largest job platform, substantially changed what people clicked and applied to, and broadened their search, but did not on average improve job-finding or earnings.

For HelloLilly: a better search engine is not the product. Matching alone does not solve unemployment, which is why the system is built around execution, people, coaches and routes, not around a ranking algorithm.

CV tools improve intermediate scores, not proven hiring

Indicative. Applicant-tracking-optimised CV tools improve keyword and structure scores, but their effect on actual hiring is under-researched.

For HelloLilly: the Application Check is honestly labelled a simulation, and CV export is tested for parseability rather than sold with a universal "ATS score".

Self-service career tools alone are the weakest option

Good, with a scope caveat. Career interventions delivered by computer alone were not significant in a meta-analysis; counsellor involvement significantly outperformed them. The evidence predates modern AI and rests on few studies, so it does not prove AI is weak, but it puts the burden of proof on AI-alone.

For HelloLilly: the coach stays in the loop by design. AI is the coach's force-multiplier, not the coach's replacement.

Digital-first widens the gap

Good. Higher digital skills make online job search more successful and less exhausting; low digital skills reduce success and raise burnout. Because digital skills follow education and income, digital-first tools risk amplifying inequality. Blended human and guided digital help is needed for low-literacy and low-access groups.

For HelloLilly: no digital gate. Spoken mode, one question at a time, plain Swedish, mobile first, and the do-it-together mode with a coach are design rules, not features.

SECTION 04

Re-skilling and Education

04

Training pays off, slowly

Strong. Labour-market training shows near-zero or negative returns while people are studying, turning positive at around two to three years. Swedish labour-market training shows a clear long-run income gain (around 8,900 kronor at three years, about 6,500 at five, persisting for roughly nine years in a study of some 90,000 participants).

Short and demand-aligned beats long

Good. Labour-market training shows about 7 per cent higher income at five years than comparable municipal vocational study, with courses about a third as long. Training passes cost-benefit tests mainly when demand is strong and targeting careful, and effects have varied strongly between decades.

Vocational college outcomes are good but cooling

Good. 81 per cent of 2024 vocational college graduates were in work about a year later, down three points and the lowest since 2013 (earlier cohorts reached 89 to 91 per cent). A descriptive share, not a causal effect.

Recognition of prior learning shortens paths

Good. Effective course choice combines labour-market data with recognition of what the person already knows (validering), avoiding redundant training. No open national data source yet shows which programme recognises what.

The grant reaches the advantaged first

Good, correlational. The transition study grant has been used disproportionately by higher-educated workers and those in occupations at low risk from AI. Higher-risk individuals apply more but are granted less, partly because a fixed budget handled first-come-first-served rewards the fastest and most complete applicants. Some funded study may have happened anyway.

Retraining does not always move people out of exposed work

Indicative, preprint, United States. A large non-peer-reviewed analysis of US public retraining found that many participants stayed in the same occupation or industry, and that wage gains may partly be catch-up. It does not transfer directly to Sweden, but it supports the same honest framing.

For HelloLilly, all of section 4: the fork that always shows applying now; labour-market training offered first; courses ranked by demand; the Prior Learning Check before any course; the Funding Navigator built to make the underserved the best prepared; honest timelines in months and years; and no promise of escape from a changing labour market. All of this is the Learn and Re-skill route.

SECTION 05

Who Is Hit, and How

05

GROUP	WHAT THE EVIDENCE SHOWS	CONFIDENCE	WHAT HELLOLILLY DOES
Long-term unemployed	Stigma, skill erosion and mental-health burden compound; training and wage subsidies show larger effects for this group	Good	Momentum early, placements, and the coach for motivation
Older jobseekers	In Sweden, callbacks fall from the early forties onwards (about half a percentage point per year), steeper for women; strongest against older women in US field experiments	Strong (causal, callbacks not hires)	Positioning strategy, employer relationships and the coach network; never "take another course" as the answer to bias
Forced changers	Forced change can leave a wellbeing deficit lasting years after re-employment; money is the most consistent predictor of how a transition ends	Good, correlational	The Transition Compass asks why first; a wellbeing check in the Transition View; Transition Economics
Migrants	Language, non-recognised credentials and discrimination; language training combined with work practice is promising; early intensive integration speeds entry but the long-run gap closes as others catch up	Good	Plain-Swedish explanations, the Prior Learning Check for foreign qualifications, placements
Neurodivergent people	Tailored one-to-one job development with on-the-job support raises employment; job carving and adapted environments help	Indicative for this group	Specialist coaches reachable through Coach Network, one step at a time, no nagging
Low digital literacy	Less success and more burnout online; blended support needed	Good	No digital gate; the do-it-together mode
Young people outside work and education	Overlapping risks; early outreach and coordinated services matter; Swedish implementation has been fragmented	Indicative	One coordinated system rather than scattered services
People on sick leave	A new instrument from March 2026 lets a person on full sickness benefit test work at their own employer for limited periods while keeping the benefit	Good (statute)	Surfaced by the Transition Compass as a matter of course

SECTION 06

How Big the Disruption Is

06

Projection. Employers in the World Economic Forum's 2025 survey project 170 million jobs created and 92 million displaced between 2025 and 2030, a structural churn of 22 per cent (creation plus displacement, not 22 per cent of jobs eliminated), and expect 39 per cent of workers' core skills to change by 2030.

Projection. The often-quoted 47 per cent of US jobs "at high risk" is an occupation-based estimate, not a prediction of losses. Task-based estimates are far lower: around 9 per cent (2016) to 14 per cent (2018) at high risk across OECD countries, with about a third facing significant change. Sweden sits near the lower end. Later work found jobs at high risk did not collapse as feared.

For HelloLilly: the disruption is real enough to justify a serious second route, and not large enough to justify fear-based selling. The system speaks about change calmly and in dated facts.

SECTION 07

From Evidence to Design

07

The table below is the bridge between research and system. Every row is a design decision in the Version 4 vision that rests on evidence above.

EVIDENCE	DESIGN DECISION	WHERE IT LIVES IN THE SYSTEM
Help works only when it builds skill **and** motivation	The division of labour: the system carries skill and execution, the coach carries motivation	The whole system; the coach section of the paper
The counsellor is the reliably effective ingredient	AI as the coach's force-multiplier; coach sign-off on big decisions	Coach tools, Transition View
Digital-first widens the gap	No digital gate as a design rule	Spoken mode, plain Swedish, mobile first, do-it-together
Unemployment causally harms mental health	Structure, contact and a kind tone; argue for the person	Progress Support, Community, all copy
Job clubs work	Peers, structure and accountability built in	Community, Expertise Exchange, Progress Support
Mastery matters most for those starting lowest	Rehearsal and accompaniment as part of the outcome	Interview Training, LinkedIn Helper, Coach Review
Untargeted volume does not work	Fewer, better applications	Match Analysis, Go or Cut, tailored CV, Contact Planner
Better recommendations do not by themselves raise job-finding	The product is execution and people, not a ranking algorithm	The routes and the human layer
Rapid placement with support beats train-then-place	Placements as a first-class route; "apply now" always shown	Placement Finder, Employer Bridge, the fork
Training pays off slowly; short and demand-aligned is better	Honest timelines; labour-market training first; demand-ranked courses	Education Matcher, Transition Economics
Recognition shortens paths	Check recognition before recommending courses	Prior Learning Check
The grant reaches the advantaged first	Make the underserved the best prepared	Funding Navigator
Forced change and money shape wellbeing	Ask why first; show the money	Transition Compass, Transition Economics
Age bias is causal and starts early	Positioning and employer relationships, not more courses	Coach Network, Network Match, Transition View
Information-only measures have limited impact	Every explanation ends in a next step	Situation Guide, Lilly
ATS tools prove intermediate scores, not hiring	Honest simulation labels; tested parseability	Application Check, CV export

SECTION 08

What a Digital System with Coaches Can Carry 08

Adapted from the July research, using the corrected evidence:

NEED	CAN THE SYSTEM PLUS COACH HELP?	HOW, IN HELLOLILLY
Low job-search self-efficacy	Strong	Accompanied execution, rehearsal, visible progress, coach
Mental distress	Mixed	Structure and contact; the coach refers onward when clinical help is needed
No target role	Mixed	Direction Explorer, with the coach to validate
Identity and sunk cost	Mixed	Stories from people who made the move; the coach for depth
Understanding the money	Strong	Funding Navigator, Transition Economics
Mapping skills to occupations	Strong	Direction Explorer on the national skills taxonomy
Knowing what is in demand	Strong	Demand data in the Education Matcher and Job Radar
Foreign credential recognition	Strong as a guide	Prior Learning Check, in plain Swedish
Age bias	Mixed	Positioning, employer relationships, coach network
Following through	Strong	One next step, reminders, if-then planning
Testing before investing	Mixed	Test Before You Invest; supply depends on employers and coach networks
Peer and role-model support	Mixed	Community and Expertise Exchange, with moderation
People excluded by digital tools	Not alone	Only with the do-it-together mode and mobile-first design
Hand-off to a human	Strong	Coaching Prompts, Lilly hand-over, Case Record

SECTION 09

What We Should Measure

09

To learn what works for whom, the research is clear that placement rates alone are not enough.

KIND	MEASURES
Outcomes	Time to work or education; employment at 6, 12 and 24 months; earnings; job quality (contract, hours, match to training); retention
Intermediate	Job-search self-efficacy, motivation, confidence, wellbeing indicators (with consent)
Process	Applications and their quality, interviews per application, routes used, support taken up, plans followed
Who	Prior work and education, recognised skills, digital comfort, constraints, always with consent and minimum five people per figure

Re-skilling must be judged on a horizon of years. A system that judges it on six to twelve months will systematically undervalue it. And because outcomes depend on many components at once, the Outcome Engine records dated facts and never claims cause.

SECTION 10

What Not to Cite

10

These figures circulate widely. They are wrong, mis-attributed, unverified or retired, and must not be repeated in any HelloLilly material.

DO NOT SAY	SAY INSTEAD
The transition grant pays "about 19,400 kronor a month"	80 per cent of income, capped at 5,773 kronor a week before tax (2026), always with the year
Vocational college graduates are "89 per cent employed"	81 per cent of 2024 graduates, the lowest since 2013; 89 per cent was the 2020 cohort
"22 per cent of jobs will be eliminated"	22 per cent structural churn (creation plus displacement); displacement alone about 8 per cent
"27 per cent of jobs are at high risk (task-based)"	Task-based estimates are 9 to 14 per cent; 27 per cent comes from a different method
JOBS II gave "55 per cent higher odds of employment"	Significantly higher re-employment, largest for those starting with low motivation
Refugee early integration gave "a lasting 15-point gain"	A short-run gain that faded as others caught up: faster integration, not a higher level
Age discrimination "accelerates near retirement"	It begins in the early forties and is steeper for women
Choosing the change is "the main predictor" of wellbeing	Money is the more consistent predictor; forced change leaves lasting effects
"Career counselling has an effect of d = 0.77"	The defensible benchmark is d = 0.35 to 0.45
Placement rates (such as "88 per cent found a job") as proof a programme worked	Placement rates describe; only controlled comparisons show cause
"Fabrication is structurally impossible" in our CV tools	Fabrication is caught by an independent check and regression-tested against an adversarial corpus

Appendix: The Swedish Data Behind the System

Verified in June 2026. The core sources are open, free and need no key; a few are files rather than live services.

NEED	SOURCE	ACCESS	NOTE
All publicly funded courses	Susa-navet (Skolverket), one national feed covering komvux, vocational college, university and labour-market training	Open, no key, daily	About 272,000 education events; verified live
Planned adult education	Skolverket Planned Educations	Open, no key	Supplementary
Occupations and skills	JobTech Taxonomy (Arbetsförmedlingen), including the European ESCO links	Open for reading	The join key for everything else
Education to occupation	JobEd Connect	Open in practice; key may be enforced later	The bridge from course content to occupations and skills
Shortage and five-year outlook	Yrkesbarometern	Open file, twice a year	To be inspected before it becomes the ranking backbone
Vacancies	SCB statistics (vacancy and recruitment tables)	Open API	Verified live
Live advert volume	JobTech JobSearch and Historical Ads	Free key or open	Well documented
Vocational college outcomes	SCB and MYH statistics	Open, partly reports	Licence terms for MYH files to be confirmed
Recognition of prior learning	No open source per programme	Gap	Guidance only; never invented

Not usable: the university admissions statistics site (no API), the credential exchange service (institution-only), and the EU job portal (no open outbound access).

Sources. Card, Kluve and Weber, active labour-market policy meta-analyses; Liu, Huang and Wang 2014; Paul and Moser 2009; Kanfer, Wanberg and Kantrowitz 2001; Vinokur and others 2000 (JOBS II); Azrin's job-club studies; Bond, Drake and Pogue 2020 and Bejerholm and others 2015 (IPS); Whiston and others 2017; IFAU reports including 2025:2 and Vikström and van den Berg 2022; Fredriksson and Seim, SNS Analys 104 (2024); CSN statistics (February 2026); MYH and SCB graduate surveys; Carlsson and Eriksson 2019; Neumark, Burn and Button 2019; Vogelsang, Olson and Shultz 2018; Dahlberg, Egebark and Vikman (IFAU 2023 and 2024); OECD (Arntz and others 2016; Nedelkoska and Quintini 2018; Bechichi and others 2019); WEF Future of Jobs 2025; Proposition 2025/26:53; Jacobs and Canedy 2026 (preprint). Full references are in the underlying research files.